Melih Barsbey

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SUMMARY

AI researcher focused on reliable and efficient machine learning under real-world constraints. My latest work investigates how the architectural pressure of compression (quantization, pruning) reshapes representation geometry, and how those geometric shifts in turn govern the robustness and safety of multimodal systems. Published in *Nature Medicine*, *NeurIPS*, *ICML*, *ICLR*, and *ICCV*. Other research highlights include interpretability in deep learning, and using human-AI complementarity and noisy supervision for medical decision-making. Industry experience in research and deployment of ML systems in various domains, as team lead, individual contributor, and freelance consultant.

EDUCATION

Boğaziçi University, Istanbul, Turkey

- Ph.D. in Computer Engineering — Sep 2016 – Sep 2023
  - Focus: Machine learning, Bayesian statistics
  - Advisors: Prof. A. Taylan Cemgil and Assoc. Prof. Arzucan Özgür
  - Cumulative GPA: 4.00 / 4.00
- M.A. in Psychology — Sep 2011 – Aug 2014
  - Focus: Embodied language processing, causal explanations
  - Thesis: Perceptual simulations beyond the sentence: the effect of narrative shifts on perceptual simulations
  - Cumulative GPA: 4.00 / 4.00
- B.A. in Psychology — Sep 2007 – Jun 2011
  - Cumulative GPA: 3.79 / 4.00

ACADEMIC AND INDUSTRY EXPERIENCE

Imperial College London, London, United Kingdom

- Post-Doctoral Research Associate in Department of Computing. — Jul 2024 – Present
  - Research focus on synergies and trade-offs between robustness and compressibility in deep learning, with applications in multiple modalities.
  - Contributions to research on mechanistic interpretability, topological deep learning for drug discovery, grokking and learning theory.
  - Publications in ICLR (x2), ICCV, ICML, Medical Image Analysis.

DeepMind, London, United Kingdom

- Research scientist internship with Robust and Verified AI team. — Jun 2022 – Oct 2022
  - Research on developing a confidence-based deferral system that leads to robust human-AI hybrid performance with sparse medical data.
  - Research on a framework for training, performance evaluation, and expert reliability analysis under noisy annotations.

Freelance Machine Learning Consultant, Istanbul, Turkey

- Freelance consultancy for multiple software companies on the research, development, and deployment of machine learning systems. — Jul 2019 – Jun 2024
  - Development and deployment of AutoML pipelines for demand forecasting.

Arute Solutions Software Consulting Inc., Istanbul, Turkey

- Senior Data Scientist, Team Lead — Sep 2017 – Jun 2019
  - Algorithm development, software design and implementation of the flagship data science product, conducting cash transaction forecasting in over 8000 ATMs.

Acıbadem University, Istanbul, Turkey

- Instructor, Department of Psychology — Nov 2014 – Sep 2017
  - Courses Taught: Descriptive and Inferential Statistics, Psycholinguistics

SELECTED PUBLICATIONS AND PREPRINTS

[17] **Barsbey, M.**, Ribeiro, A. H., Şimşekli, U., & Birdal, T. On the Interaction of Compressibility and Adversarial Robustness. *International Conference on Learning Representations (ICLR)*, 2026.

[16] Prieto, L., Stevinson, E., **Barsbey, M.**, Birdal, T., & Mediano, P. A. M. Correlations in the Data Lead to Semantically Rich Feature Geometry Under Superposition. *International Conference on Learning Representations (ICLR)*, 2026.

[15] **Barsbey, M.**, Prieto, L., Zafeiriou, S., & Birdal, T. Large Learning Rates Simultaneously Achieve Robustness to Spurious Correlations and Compressibility. *International Conference on Computer Vision (ICCV)*, 2025.

[14] Prieto, L., **Barsbey, M.**, Mediano, P. A. M., & Birdal, T. Grokking at the Edge of Numerical Stability. *International Conference on Learning Representations (ICLR)*, 2025.

[13] Stutz, D.[*], Cemgil, A.T.[*], Roy, A.G.[*], Matejovicova, T.[*], **Barsbey, M.[*]**, et al. Evaluating AI Systems under Uncertain Ground Truth: A Case Study in Dermatology. *Medical Image Analysis*, *103*, p.103556, 2025. [*]Equal contribution.

[12] Wan, Y.[*], **Barsbey, M.[*]**, Zaidi, A., & Şimşekli, U. Implicit Compressibility of Overparametrized Neural Networks Trained with Heavy-Tailed SGD. *International Conference on Machine Learning (ICML)*, 2024. [*]Equal contribution

[11] Dvijotham, K.[*], Winkens, J.[*], **Barsbey, M.[*]**, Ghaisas, S.[*], Pawlowski, N. et al. Enhancing the reliability and accuracy of AI-enabled diagnosis via complementarity-driven deferral to clinicians (CoDoC). *Nature Medicine*, 29 (7), 1814-1820, 2023. [*]Equal contribution.

[10] **Barsbey, M.**, & Cemgil, A. T. Modeling Hierarchical Seasonality through Low-Rank Tensor Decompositions in Time Series Analysis. *IEEE Access*, 11, 85770-85784, 2023.

[9] **Barsbey, M.[*]**, Özçelik, R.[*], Bağ, A. et al. pydebiaseddta: A Computational Software for Training Robust Drug-Target Affinity Prediction Models. *Journal of Computational Biology*, 30 (11), 1240-1245, 2023. [*]Equal contribution.

[8] **Barsbey, M.** *Utilizing Nonnegative Tensor Factorization Methods for Inference, Model Selection, and Analysis in Supervised Learning*. PhD dissertation. Boğaziçi University, Istanbul, Turkey, Sep 2023.

[7] Raj, A., **Barsbey, M.**, Gürbüzbalaban, M., Zhu, L., & Şimşekli, U. Algorithmic Stability of Heavy-Tailed Stochastic Gradient Descent on Least Squares. *Conference on Algorithmic Learning Theory (ALT)*, 2023.

[6] **Barsbey, M.[*]**, Sefidgaran, M.[*], Erdogdu, M. A., Richard, G., & Şimşekli, U. Heavy Tails in SGD and Compressibility of Overparametrized Neural Networks. *Advances in Neural Information Processing Systems Conference (NeurIPS)*, 2021. [*]Equal contribution

[5] Yıldırım, S., Kurutmaz, M. B., **Barsbey, M.**, Şimşekli, U., & Cemgil, A. T. Bayesian Allocation Model: Marginal Likelihood-based Model Selection for Count Tensors. *IEEE Journal of Selected Topics in Signal Processing*, 15(3), 560-573, 2021.

[4] Kurutmaz, M. B.[*], **Barsbey, M.[*]**, Cemgil, A. T., Yıldırım, S., & Şimşekli, U. Bayesian Model Selection for Identifying Markov Equivalent Causal Graphs. Paper presented at *Symposium on Advances in Approximate Bayesian Inference (AABI)*, 2019. *Equal contribution

[3] **Barsbey, M.** *Perceptual Simulations beyond the Sentence: The Effect of Narrative Shifts on Perceptual Simulations*. Master's thesis. Boğaziçi University, Istanbul, Turkey, Aug 2014.

[2] **Barsbey, M.**, Sarıbay, S. A., Boduroğlu, A., & Zwaan, R. A. Perceptual simulations beyond the sentence: the effect of narrative shifts on perceptual simulations. Poster presented at the *Annual Conference on Embodied and Situated Language Processing*, Rotterdam, The Netherlands, 2014.

[1] **Barsbey, M.**, & Lombrozo, T., Linguistic differences in explanation requests and their effects on the evaluation of explanations: the case of English and Turkish. Poster presented at the annual meeting of the *Cognitive Science Society*, Berlin, Germany, 2013.

SCIENTIFIC SERVICE

Reviewer

- Neural Information Processing Systems (NeurIPS) 2019 – Present
- International Conference on Machine Learning (ICML) 2021 – Present
- International Conference on Learning Representations (ICLR) 2021 – Present
- Transactions on Machine Learning Research (TMLR) 2023 – Present
- Conference on Computer Vision and Pattern Recognition (CVPR) 2024 – Present

AWARDS & SCHOLARSHIPS

- Global Talent Visa - United Kingdom 2024 – 2029
- PhD Scholarship from The Scientific and Technological Research Council of Turkey (TÜBİTAK) 2016 – 2021